Where You Refuel

Dataset description, exploratory analysis and data quality assessment

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Data Visualization · Technical report

1. Scope and purpose

This report documents the data behind a single question - *how much does where you refuel change what you pay?* - restricted to the 22 provinces of ISTAT NUTS-1 **Nord-Est** (Trentino-Alto Adige, Veneto, Friuli-Venezia Giulia, Emilia-Romagna) over **2024–2025**, sampled **one day per week (Monday)**.

The restriction is deliberate. The question is geographic, not dynamic, so daily resolution buys nothing; and, as Section 5.3 shows, the 2024–2025 window is internally consistent in a way the full 2015–2026 archive is not.

Headline: 105 weekly extracts yield **2,000,810 usable price observations** (97.14% of the Nord-Est rows) across **5,329 registered stations**, with only 125 rows discarded for quality reasons. The dataset is in good condition; its real difficulties are semantic (free-text fuel names, mixed units) rather than structural.

Sections 2–5 describe the data and assess its quality. Section 6 reports what exploration established about whether the data can carry the intended visualization, and which of its properties constrain the design.

2. Sources and provenance

Source	Content	Licence
MIMIT *Carburanti - archivio storico prezzi*	daily station prices, quarterly `.tar.gz`, 2015 Q1–	IODL 2.0
MIMIT *Carburanti - anagrafica impianti attivi*	station registry with coordinates	IODL 2.0
openpolis/geojson-italy	province boundaries (ISTAT-derived)	CC-BY

Prices are reported to the Ministry by operators and each daily file is an extract of the prices valid at 08:00. Archive URLs follow `opendatacarburanti.mise.gov.it/categorized/{dataset}/{year}/{year}_{q}_tr.tar.gz`.

3. Structure of the raw data

Prices (`prezzo_alle_8-YYYYMMDD.csv`) - line 1 is a free-text extraction stamp, line 2 the header, delimiter `;`:

Field	Type	Notes
`idImpianto`	int	joins to the registry
`descCarburante`	text	**free text**, see §5.4
`prezzo`	float	EUR/litre, or EUR/kg for methane and LNG
`isSelf`	0/1	1 = self-service, 0 = attended
`dtComu`	datetime	when the operator filed this price

Registry (`anagrafica_impianti_attivi-YYYYMMDD.csv`) - same two-line preamble, delimiter `;`, ten fields: `idImpianto`, `Gestore`, `Bandiera`, `Tipo Impianto`, `Nome Impianto`, `Indirizzo`, `Comune`, `Provincia`, `Latitudine`, `Longitudine`. `Tipo Impianto` takes two values, `Stradale` or `Autostradale` - a controlled vocabulary the parser relies on (§5.5).

Although the registry is published every day alongside the price file, the station population changes relatively slowly. We therefore retained eight registry snapshots, one per quarter. The number of national registry records increased from 22,821 to 24,462 over the two-year period.

4. Construction of the analysis panel

The pipeline (`scripts/`, ~600 lines of Python) runs in four stages:

1. **`geo.load_registry`** - parse all eight snapshots, restrict to Nord-Est, derive region, motorway flag, harmonised brand, and a geocode validity flag; take the union across snapshots keeping the most recent record per station.
2. **`build_panel`** - read the 105 weekly price files, join the registry, harmonise fuel labels, apply the documented filters, and write the panel plus a machine-readable QC record (`data/processed/qc.json`).
3. **`analyse`** - compute the four geographic views.
4. **`figures`** - render the figures directly from the processed panel, quality-control records and aggregates produced by analyse, so that no number in this report is transcribed by hand.

Two methodological rules are applied throughout, because both guard against composition effects that would otherwise be mistaken for geography:

- Statistics are computed **within a week** and then averaged across weeks. Averaging raw prices over two years would let a province that happens to report more often during expensive weeks look expensive.
- Paired comparisons (the attended premium) are made **within a station**, so the gap cannot be an artefact of *which* stations offer attended service.

5. Data quality assessment

5.1 Coverage

All 105 expected Mondays are present; none is missing. Between 4,251 and 4,384 Nord-Est stations post at least one price in any given week, out of 5,329 registered. Over the whole period **5,018 stations report at least once**, so **311 registered stations (5.8%) are never observed quoting a price** - they are listed as active but silent. This is a real limitation: the "provincial mean" is a mean over stations that report, not over stations that exist.

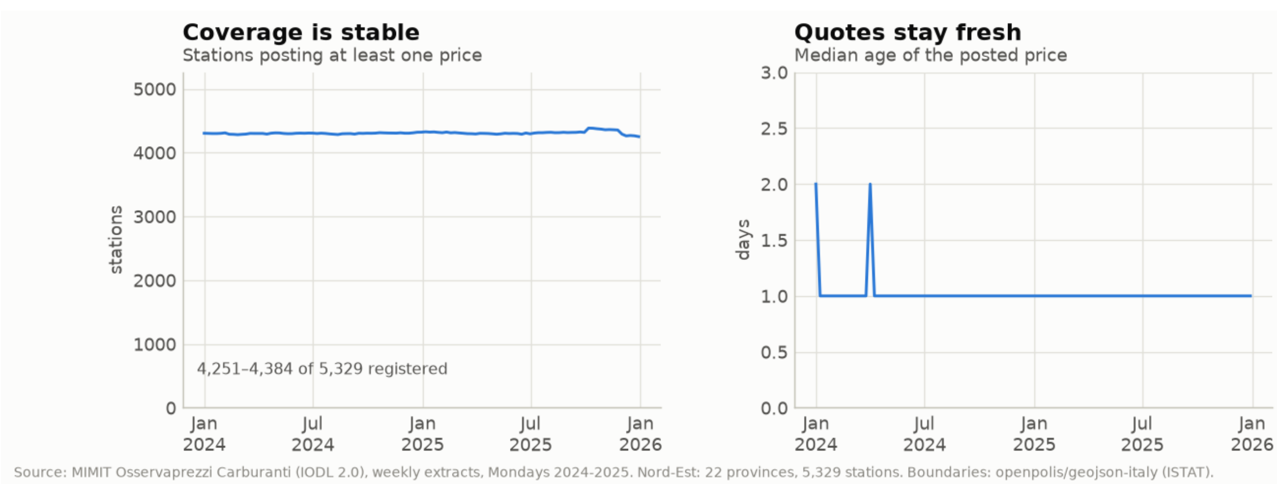

Figure 1: Weekly coverage and the freshness of quotes.

5.2 Price plausibility

Only **124 rows (0.006%)** fall outside a plausibility band of 0.5–4.0 EUR/litre and are removed; one further row has an unparseable station id. Nothing is missing: `prezzo` parses for every remaining row. The retained distribution is well-behaved, with pronounced spikes at round numbers (1.700, 1.800, 1.900) that are a genuine pricing behaviour rather than a defect.

For context, the *live* feed downloaded on 2026-08-06 carries nine such values, among them 0.100 (four stations) and 8.888 (two) - placeholders rather than prices. The band is set to catch exactly these without touching any real fuel price.

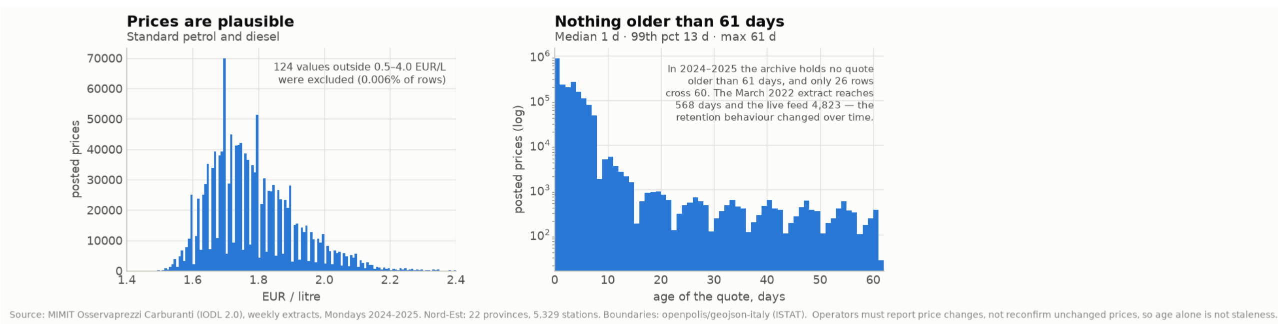

Figure 2: Price plausibility and the age of quotes.

5.3 Quote age - and a regime change in the archive

Operators are required to report price *changes*, not to reconfirm unchanged prices, so an old `dtComu` is **not by itself evidence of a stale record**. We therefore flag at 60 days and drop only beyond 365.

In the considered window the **drop** threshold never fires: no quote is anywhere near a year old, so not one row is removed for staleness. The **flag** does fire, but barely - median quote age is **1 day**, the 99th percentile 13 days, and the oldest quote in the panel is **61 days**, so **26 rows (0.001%)** cross the 60-day line. They are marked, not dropped.

That freshness is a property of *the considered window*, not of the dataset. The same measurement on an extract from March 2022 gives a different picture; since the two differ greatly in size, the last column gives the share as well as the count:

Extract	rows	median	99th pct	max age	> 60 d
Archive, 2022-03-17	95,104	6 d	453 d	568 d	6,201 (6.52%)
This panel, 2024–2025	2,000,810	1 d	13 d	61 d	26 (0.001%)

The first row is a single national day; the second is the whole panel, 105 Mondays of Nord-Est prices.

The two behave like different datasets. In 2022 a fifteenth of all quotes were over two months old and the 99th percentile sat beyond a year, so staleness was a genuine problem an analysis had to handle. By 2024 that tail has gone: the same threshold catches 26 rows in two million. **Any project spanning the full archive would have to model that discontinuity;** scoping to 2024–2025 avoids it, and that is part of why the window was chosen.

5.4 Fuel labels: the main semantic problem

`descCarburante` is free text. Across the study period it takes **59 distinct values** (56 after case-folding) for what are really **nine products**. Diesel alone appears under 20 commercial names - `Blue Diesel`, `Hi-Q Diesel`, `Supreme Diesel`, `Excellium Diesel`, `Gasolio Oro Diesel`, and so on.

Harmonisation is an explicit lookup table (`fuels.py`), not a fuzzy rule, so an unrecognised label surfaces as an error rather than being silently absorbed. After harmonisation **zero rows are unmapped**.

Two decisions matter for the analysis:

- **Premium grades are kept separate from standard ones.** Pooling them would make a station that happens to sell Blue Diesel look expensive when it is simply selling a different product.
- **Winter-grade diesel** (`Gasolio artico`, `Gasolio Alpino`, ...) is separated too: it is sold mainly in Alpine provinces and priced above standard diesel, so pooling it would bias precisely the provinces being compared.

Mixed units. Methane, LNG and L-GNC are priced **per kilogram**, not per litre. These **58,784 rows (2.854%)** are excluded from every EUR/litre view. This is the single largest exclusion in the pipeline, and it is a unit incompatibility rather than a quality failure.

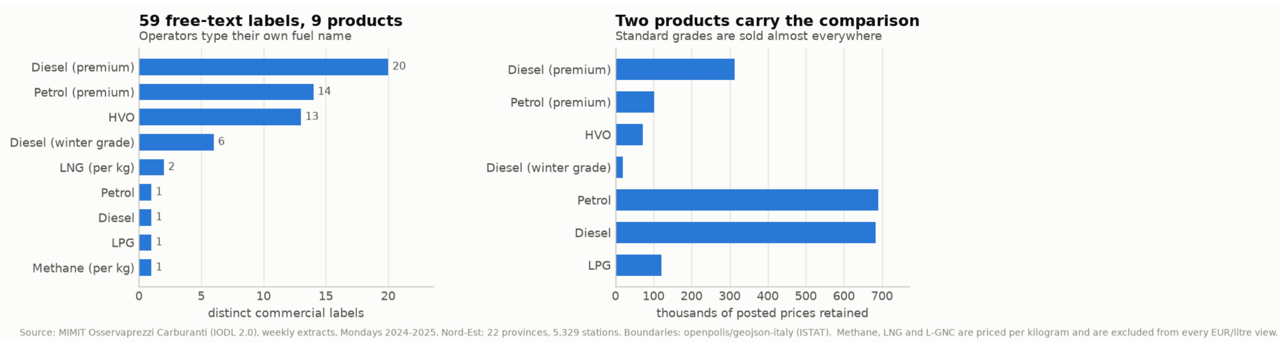

Figure 3: Label harmonisation and the products that carry the comparison.

5.5 Registry quality

- **Embedded separators.** 34 rows across the eight snapshots carry a stray `;` inside a free-text field, so they split into 11 or 12 pieces instead of 10. They are *recoverable* rather than droppable, because three positions in the row can be trusted: `idImpianto` is always first, the coordinates are always last, and `Tipo Impianto` is a two-value controlled vocabulary. We repair them rather than drop them; **zero rows are unrecoverable.** The surplus pieces are rejoined with `;`, the exact inverse of the split, so the repaired field is the ministry's original text rather than an approximation of it.
- **Geocoding.** Only **10 stations (0.19%)** have coordinates that fall outside a Nord-Est bounding box or sit at (0,0).
- **Churn.** 4,267 of 5,329 stations appear in all eight snapshots; 225 appear in only one. Because a single snapshot cannot resolve every id, the registry is built as a union across snapshots - using only the latest snapshot would have silently dropped historical stations.

5.6 Exclusion ledger

Filter	Rows	% of Nord-Est rows
Unparseable station id	1	0.000
Price missing	0	0.000
Price outside 0.5–4.0 EUR/L	124	0.006
Fuel label unmapped	0	0.000
Priced per kilogram (methane, LNG)	58,784	2.854
Service flag unknown	0	0.000
Quote older than 365 days	0	0.000
Retained	**2,000,810**	**97.14**

5.7 Why the window stops at 2024–2025

The archive reaches back to 2015. Restricting to two years is partly a matter of proportion - the question is geographic, so daily resolution over a decade adds volume rather than evidence - but the decisive reasons are properties of the older data, each checked against a 2022 file and archive held for comparison.

- **The retention regime changes (§5.3).** Maximum quote age is 61 days in 2024–2025, 568 days in the March 2022 extract. Pooling the two would mix populations whose staleness differs by an order of magnitude.
- **The archive layout changes.** 2022 quarterly archives store the daily CSVs at the root; 2024–2025 archives nest them inside a `{year}_{q}_tr/` directory. A loader written against one silently extracts nothing from the other.
- **The fuel vocabulary is not stable.** The March 2022 file uses `SSP98`, a label that never appears in 2024–2025. Because harmonisation uses a strict lookup table, every additional year requires a new pass of manual label review. Unrecognised labels are recorded in the quality-control output and excluded until an explicit mapping is provided.
- **Daily completeness is better.** 2022 Q1 is missing two consecutive days (15–16 March) out of 90. Across all of 2024–2025 exactly one day is absent from the archive, Saturday 2025-01-11 - and being a Saturday it does not touch the Monday sample, which is why all 105 expected weeks are present (§5.1).

None of this makes the earlier data unusable. It means a longer window is a different project, with a discontinuity to model rather than a panel to describe, and the two years chosen are the ones that behave consistently.

6. Exploratory analysis

Exploration had one job: establish whether this panel can carry the intended visualization, and identify the properties that constrain how it must be built. All results below use **standard self-service grades** unless stated otherwise.

6.1 Panel composition: how much data sits in each cell

Of 5,329 registered stations, **5,018 report at least once** and 5,008 of those also carry usable coordinates. They divide unevenly:

Region	Reporting stations	Product	Rows retained
Veneto	2,109	Petrol	689,555
Emilia-Romagna	1,930	Diesel	683,971
Friuli-Venezia Giulia	555	Diesel (premium)	313,321
Trentino-Alto Adige	424	LPG	121,196
		Petrol (premium)	102,031
of which motorway	*101*	HVO	71,491
		Diesel (winter grade)	19,245

Per province the station count runs from **40 (Trieste)** and 53 (Gorizia) to **392 (Verona)**, median 195. Standard petrol and diesel are the only two products stocked at nearly every station, and therefore the only two comparable across 5,000 pumps; the remainder are too unevenly distributed to map.

Three consequences for the design follow directly:

1. **A province × motorway breakdown is not supportable.** 101 motorway stations across 22 provinces is under five per province, and several provinces have none. Motorway can only be a Nord-Est-level comparison.
2. **Per-province views must expose their sample size.** A price distribution for Trieste rests on 40 pumps against Verona's 392; drawn identically, the two would imply equal confidence.
3. **Unbalanced comparisons need medians.** With 101 stations against ~4,900, a handful of outlying motorway sites would otherwise drive the result.

6.2 The intended comparisons are separable in this data

Comparison	Gap	Measured over
Attended vs self-service	13.0 c/L	240,347 paired station-weeks, 2,785 stations
Motorway vs ordinary road	10.4 c/L	101 vs ~4,900 stations, weekly medians
Spread within a province-week	9.1 c/L	p10–p90, 22 provinces × 105 weeks
Cheapest vs dearest province	8.8 c/L	RO €1.745 to BZ €1.833

Each is large relative to week-to-week movement and stable across the two years - the motorway premium, for instance, never falls below 7.8 cents in any of the 105 weeks, and the provincial ordering is coherent rather than noisy (Alpine and border provinces dear, the Po plain cheap). Diesel behaves like petrol throughout: an 11.6-cent motorway premium and a 9.8-cent provincial range. The question is answerable with this data, and the answer does not depend on which of the two headline fuels is chosen.

6.3 Two distributional properties the visualization must respect

Stations fall into two camps on attended service, so **the average describes almost none of them**. At **In 32% of paired station-weeks, the two prices differ by less than 0.5 cents per litre** and are treated as effectively identical. The rest charge **19.0 cents** more on average. The overall figure of 13.0 cents sits in the gap between the two groups and describes neither, so this comparison needs to be drawn as **a distribution rather than a single bar**.

Within-province spread is as wide as the between-province range. Inside a single province in a single week, the 10th–90th percentile of station prices spans **9.1 cents** on average, against **8.8 cents** between the cheapest and dearest province. Bologna is the most dispersed (11.8c), Bolzano the least (6.0c) - Bolzano is uniformly expensive, whereas Bologna contains both cheap and dear pumps.

This is the most consequential thing the exploration turned up. A choropleth on its own would leave a reader with “my province is expensive”, when the data supports “my province contains both the cheapest and the dearest pump I could reach”. **A map alone cannot carry this dataset**; a within-province view has to sit beside it.

7. Limitations

1. **Posted, not transacted, and unweighted.** Prices are what operators advertise, not what drivers pay, and every station counts equally regardless of throughput. Provincial means are means over pumps, not over litres.
2. **Silent stations.** 311 registered stations never report; if silence correlates with price, provincial means are biased.
3. **Friuli-Venezia Giulia** operates a regional discount for residents that posted prices do not reflect, so UD, GO, TS and PN overstate what a resident actually pays.
4. **Weekly Monday sampling** is appropriate for a geographic question but would be inadequate for a dynamic one.
5. **The clean age profile does not generalise** beyond 2024–2025 (§5.3).

8. Reproducibility

```
scripts/config.py        scope, paths, thresholds
scripts/fuels.py         59 labels -> 9 products; per-kg products flagged
scripts/geo.py           registry parsing, repair, geocode QC, boundaries
scripts/build_panel.py   105 weekly files -> panel + qc.json
scripts/analyse.py       the four geographic views
scripts/figures.py       the figures in this report
```

Running `build_panel.py`, `analyse.py` and `figures.py` in that order regenerates every number and figure in this report from the raw CSVs. Every threshold is a named constant in `config.py`; every exclusion is counted into `data/processed/qc.json` rather than applied silently. Environment: Python 3.12, pandas 3.0.5, matplotlib 3.11.1, numpy 2.5.1, pyarrow 24.0.0.